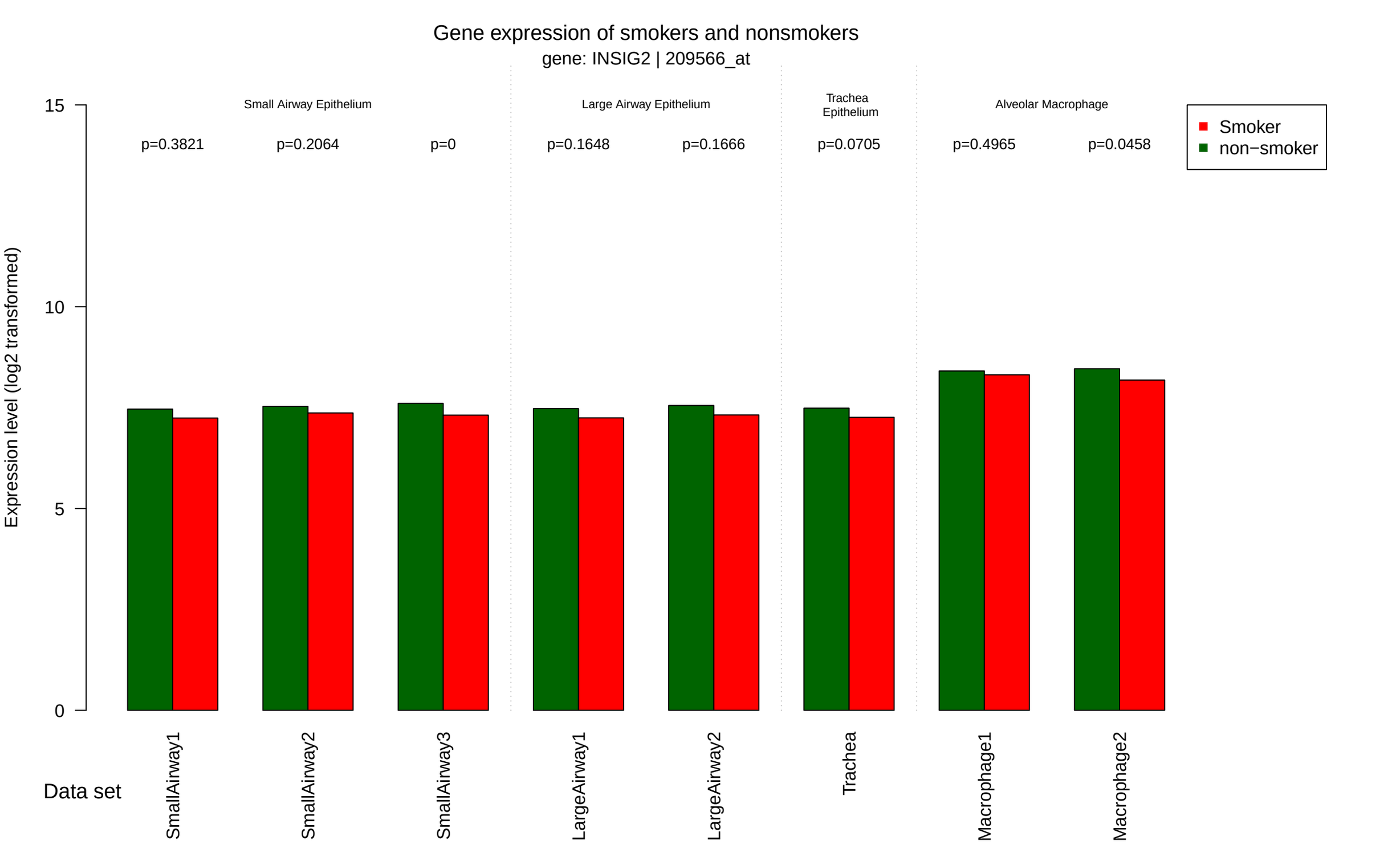
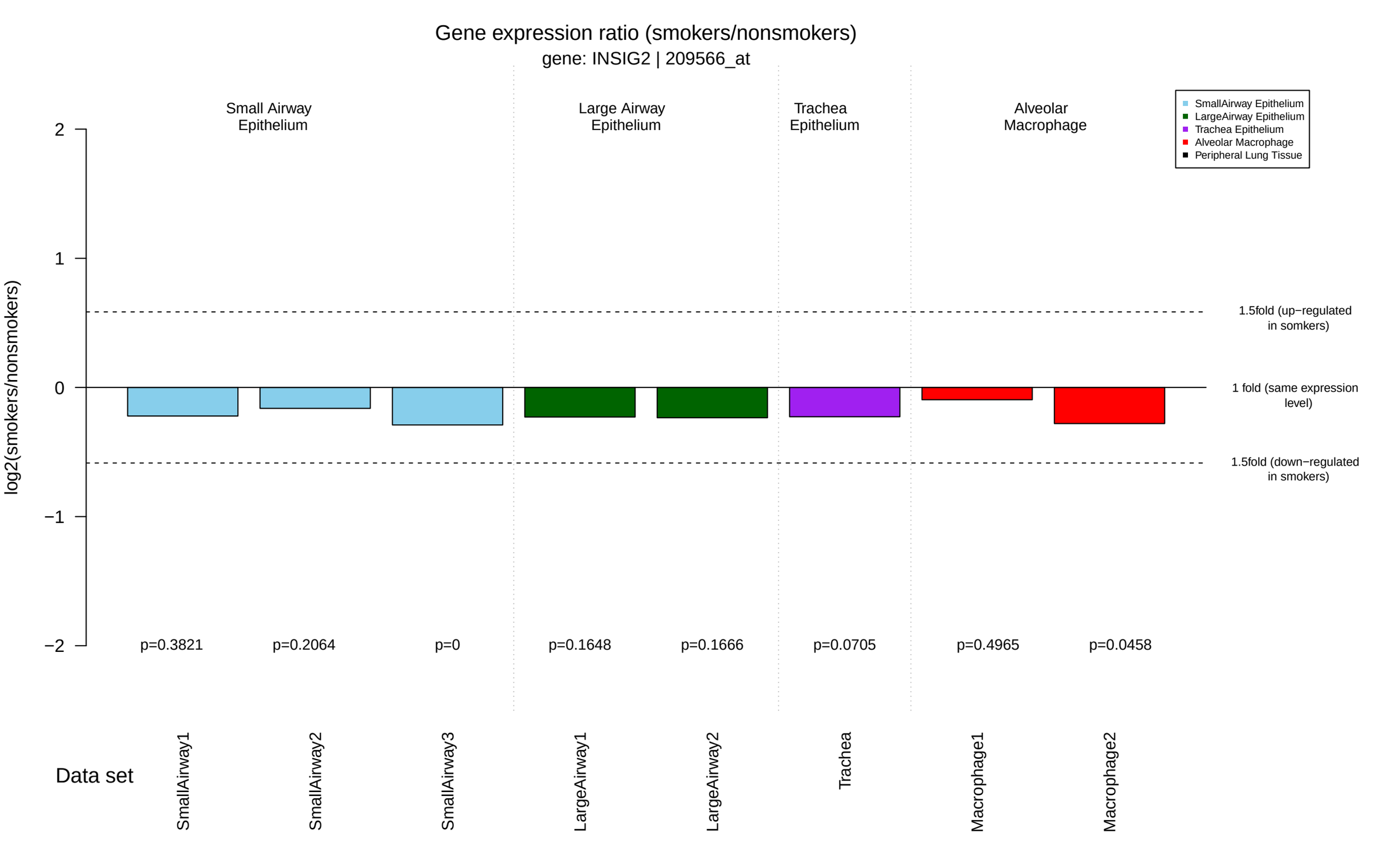
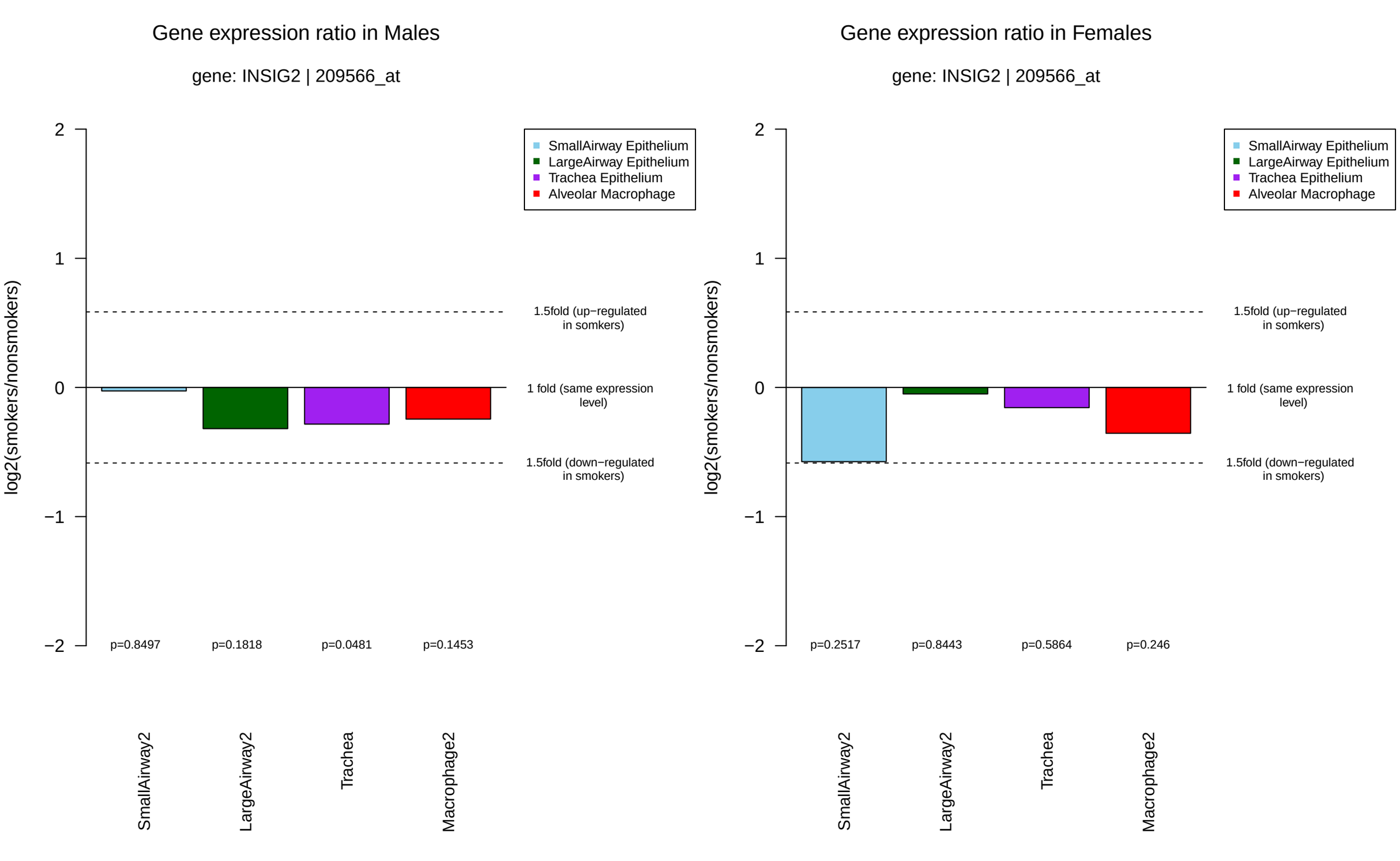
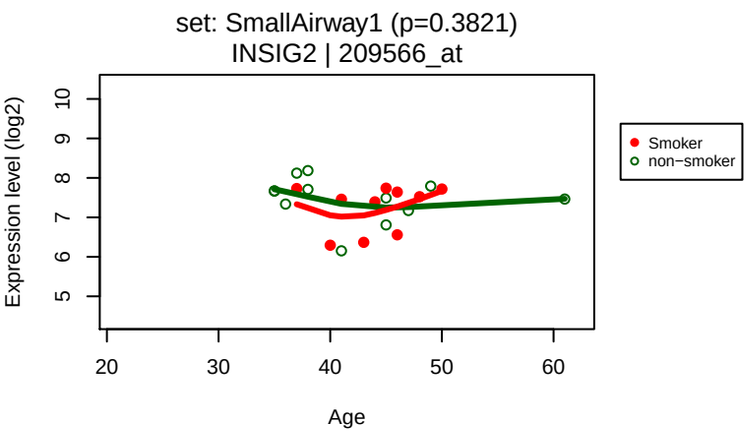
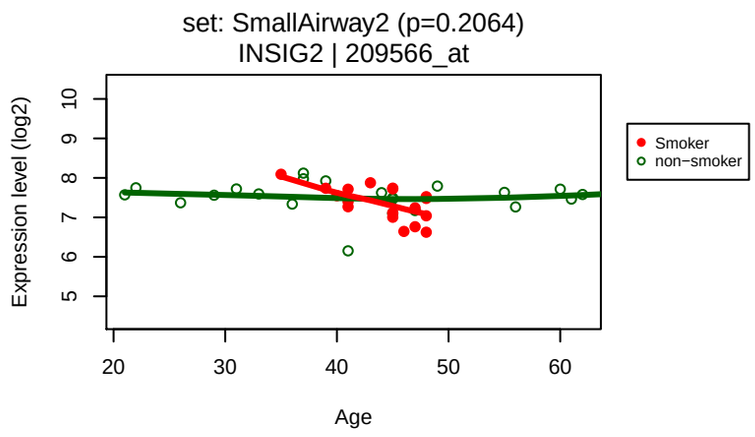
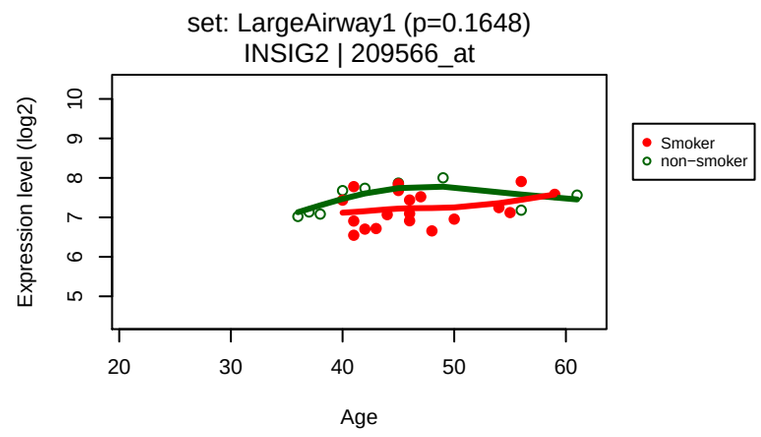

Gene expression of smokers and nonsmokers
gene: INSIG2 | 209566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3821
p=0.2064
p=0
p=0.1648
p=0.1666
p=0.0705
p=0.4965
p=0.0458
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

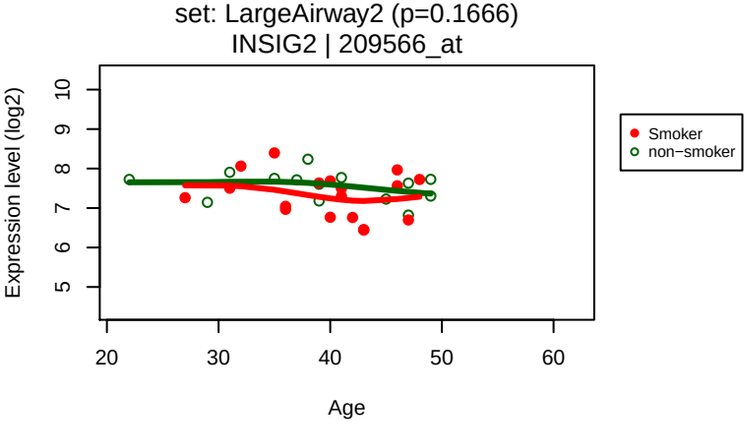

Gene expression ratio (smokers/nonsmokers)
gene: INSIG2 | 209566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3821
p=0.2064
p=0
p=0.1648
p=0.1666
p=0.0705
p=0.4965
p=0.0458
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

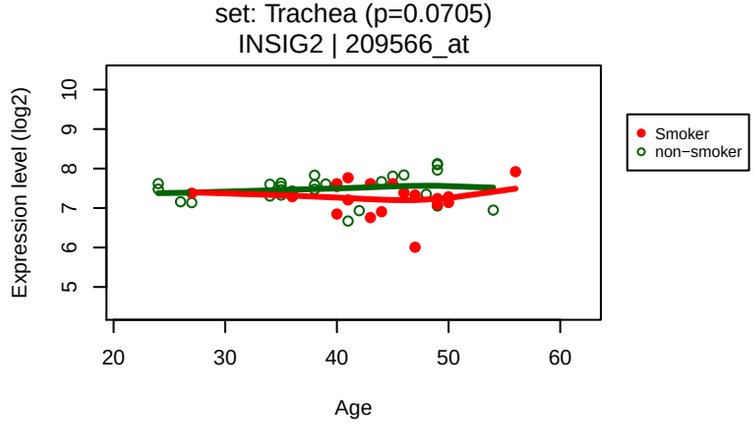

Gene expression ratio in Males
gene: INSIG2 | 209566_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8497
p=0.1818
p=0.0481
p=0.1453
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: INSIG2 | 209566_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2517
p=0.8443
p=0.5864
p=0.246
SmallAirway2
LargeAirway2
Trachea
Macrophage2

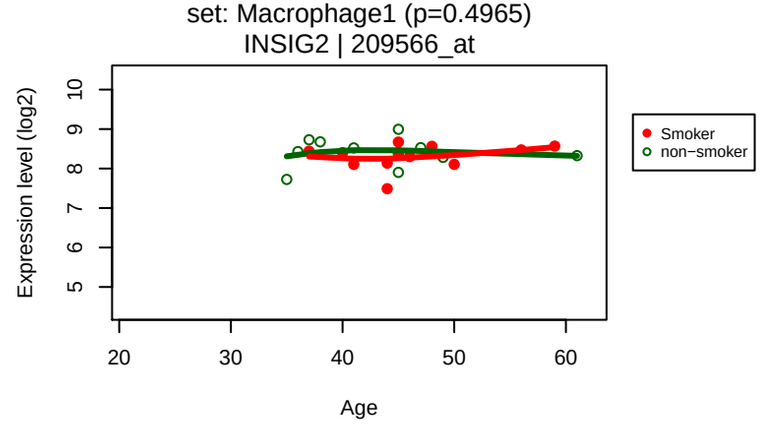

set: SmallAirway1 (p=0.3821)
INSIG2 | 209566_at
Expression level (log2)
Age
Smoker
non−smoker

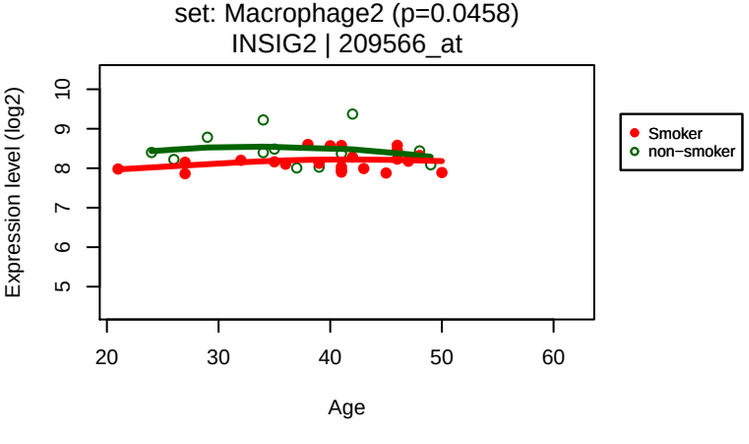

set: SmallAirway2 (p=0.2064)
INSIG2 | 209566_at
Expression level (log2)
Age
Smoker
non−smoker

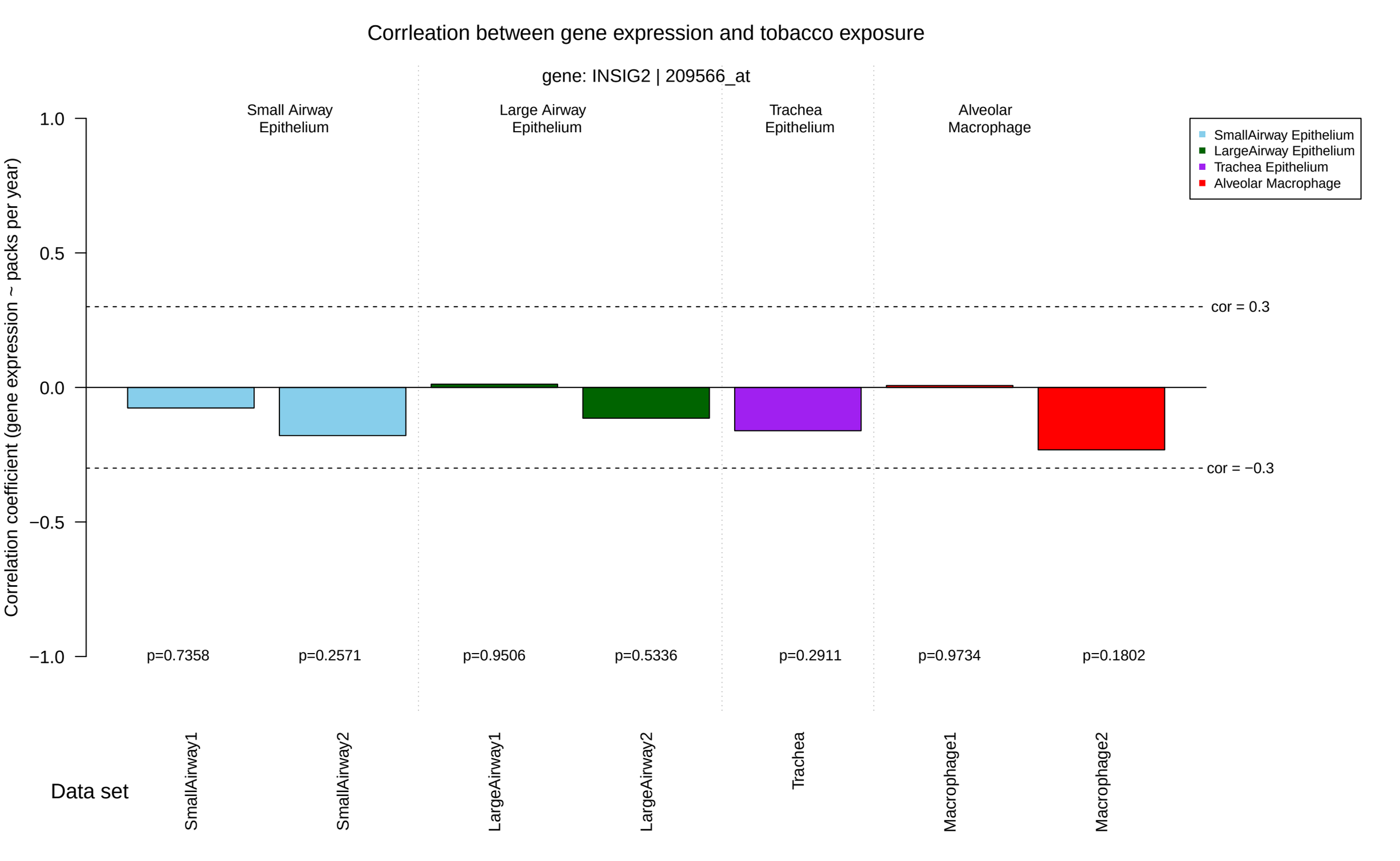

set: LargeAirway1 (p=0.1648)
INSIG2 | 209566_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1666)
INSIG2 | 209566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0705)
INSIG2 | 209566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4965)
INSIG2 | 209566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0458)
INSIG2 | 209566_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: INSIG2 | 209566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7358
p=0.2571
p=0.9506
p=0.5336
p=0.2911
p=0.9734
p=0.1802
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2